

LAB NO.01

```
import numpy as np

# Create NumPy arrays
arr1 = np.array([1, 2, 3, 4, 5])
arr2 = np.arange(10)
arr3 = np.linspace(0, 1, 5)

# Array operations
sum_arr = arr1 + 5
mean_val = np.mean(arr1)

print("Array:", arr1)
print("Mean:", mean_val)
```

 Array: [1 2 3 4 5]
Mean: 3.0

```
import pandas as pd

data = {
    'Name': ['Amit', 'Priya', 'Raj'],
    'Marks': [85, 90, 78]
}
df = pd.DataFrame(data)

print(df)
print("Average Marks:", df['Marks'].mean())
```



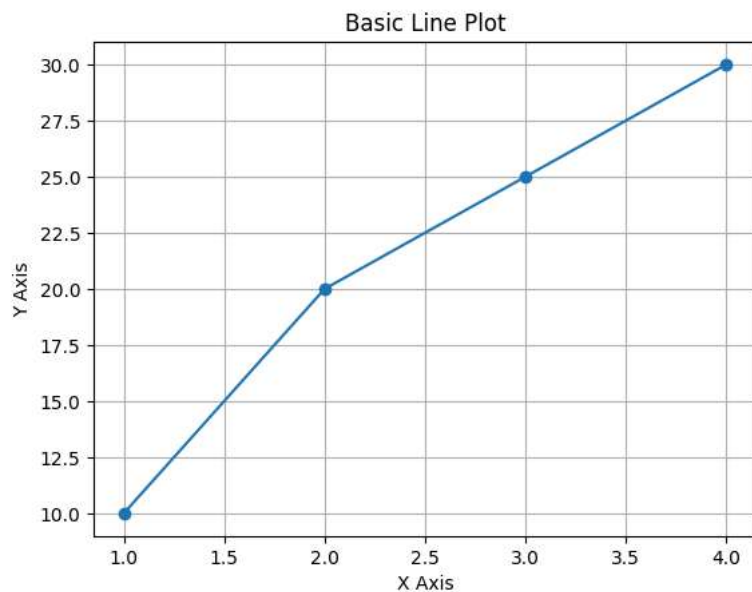
	Name	Marks
0	Amit	85
1	Priya	90
2	Raj	78

Average Marks: 84.33333333333333

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [10, 20, 25, 30]

plt.plot(x, y, marker='o')
plt.title("Basic Line Plot")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.grid(True)
plt.show()
```



LAB NO.02

```
import matplotlib.pyplot as plt
import pandas as pd

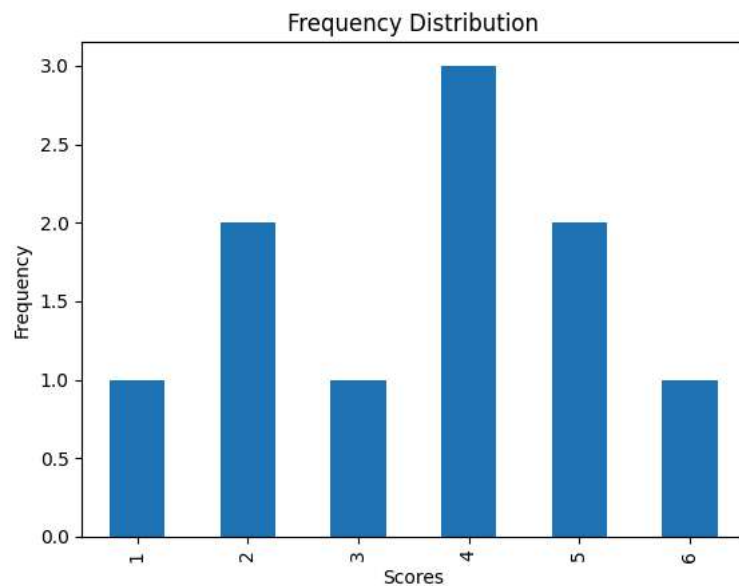
data = [1, 2, 2, 3, 4, 4, 4, 5, 5, 6]
df = pd.DataFrame(data, columns=['Scores'])

freq = df['Scores'].value_counts().sort_index()
print("Frequency:\n", freq)

freq.plot(kind='bar')
plt.title("Frequency Distribution")
plt.xlabel("Scores")
plt.ylabel("Frequency")
plt.show()
```



Frequency:
Scores
1 1
2 2
3 1
4 3
5 2
6 1
Name: count, dtype: int64



```
import numpy as np

data = [10, 12, 23, 23, 16, 23, 21, 16]

variance = np.var(data)
std_dev = np.std(data)

print("Variance:", variance)
print("Standard Deviation:", std_dev)
```

↗ Variance: 24.0
Standard Deviation: 4.898979485566356

LAB NO.03

```
import numpy as np
import statistics as stats

data = [10, 20, 30, 40, 50, 50]

mean = np.mean(data)
median = np.median(data)
mode = stats.mode(data)

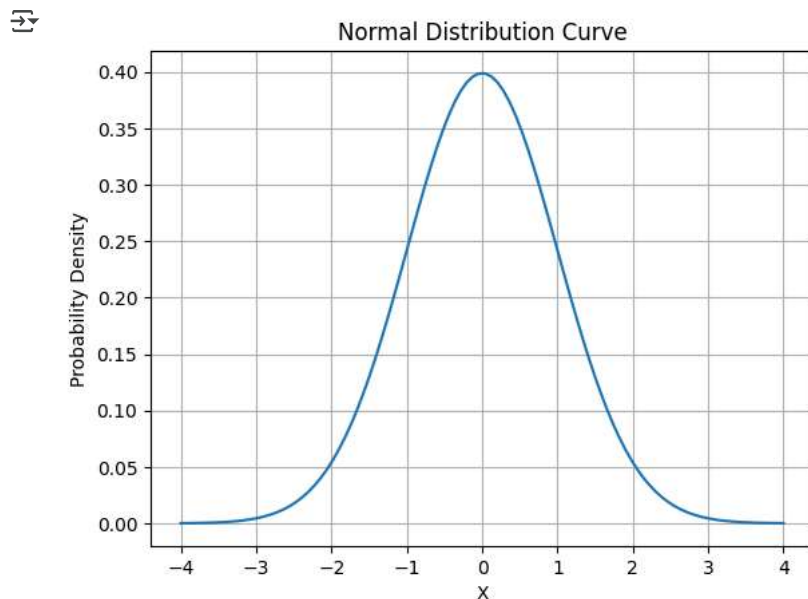
print("Mean:", mean)
print("Median:", median)
print("Mode:", mode)
```

↗ Mean: 33.333333333333336
Median: 35.0
Mode: 50

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

x = np.linspace(-4, 4, 100)
y = norm.pdf(x, 0, 1)

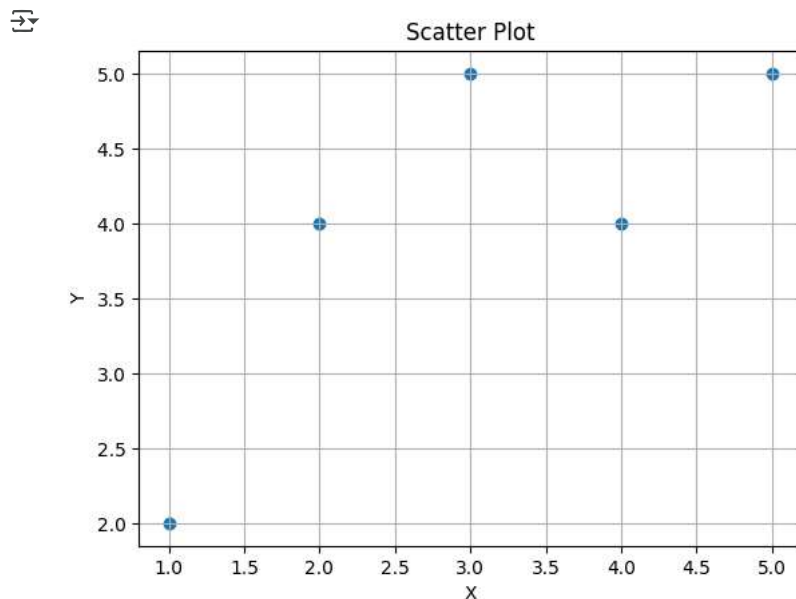
plt.plot(x, y)
plt.title("Normal Distribution Curve")
plt.xlabel("X")
plt.ylabel("Probability Density")
plt.grid(True)
plt.show()
```



```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'X': [1, 2, 3, 4, 5],
    'Y': [2, 4, 5, 4, 5]
}
df = pd.DataFrame(data)

plt.scatter(df['X'], df['Y'])
plt.title("Scatter Plot")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()
```



LAB NO.04

```
import numpy as np

x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 4, 5])

correlation = np.corrcoef(x, y)[0, 1]
print("Correlation Coefficient:", correlation)
```

Correlation Coefficient: 0.7745966692414834

```
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

x = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y = np.array([1, 2, 1.3, 3.75, 2.25])

model = LinearRegression()
model.fit(x, y)

y_pred = model.predict(x)

plt.scatter(x, y, color='blue')
plt.plot(x, y_pred, color='red')
plt.title("Simple Linear Regression")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()
```

```
print("Slope:", model.coef_[0])  
print("Intercept:", model.intercept_)
```

